
CREATIVITY PARITY, DIVERSITY LOSS: AI-GENERATED STORIES MATCH HUMAN QUALITY BUT EXHIBIT 63% LESS STYLISTIC VARIATION

[ANONYMIZED FOR REVIEW]

ABSTRACT

Objective: Generative AI tools produce creative text that rivals human output in perceived quality, but whether this individual parity comes at the cost of reduced collective diversity remains poorly understood. We tested whether AI-generated short stories are rated as comparably creative to human-authored stories and whether AI output exhibits homogenization at the corpus level. **Method:** In a preregistered between-subjects evaluation design, 380 German-speaking participants ($M_{\text{age}} = 34.5$ years, $SD = 11.8$; 53.7% male) each read one short story—either human-authored ($n = 195$ evaluations of 98 stories) or AI-generated by ChatGPT ($n = 185$ evaluations of 96 stories)—and rated it on perceived novelty ($\alpha = .92$), enjoyment ($\alpha = .89$), and narrative transportation ($\alpha = .83$) using 7-point Likert scales. **Results:** AI-generated stories were not rated as more novel than human-authored stories, $t(378) = 0.61$, $p = .544$, $d = 0.06$, 95% CI $[-0.14, 0.26]$. However, AI stories showed a 63.4% reduction in collective diversity—operationalized as 1 minus mean pairwise cosine similarity of LIWC-22 linguistic feature vectors—compared to human stories, permutation $p = .0001$. Human-authored stories were rated as more narratively transporting, $t(378) = -2.03$, $p = .043$, $d = -0.21$. **Conclusion:** AI-generated creative output achieves perceived quality parity with human work but produces a substantially narrower range of stylistic expression, raising concerns about the long-term cultural consequences of widespread AI adoption in creative domains.

Keywords generative AI; creativity; collective diversity; narrative transportation; computational linguistics

Creativity Parity, Diversity Loss: AI-Generated Stories Match Human Quality but Exhibit 63% Less Stylistic Variation

The rapid diffusion of generative artificial intelligence (AI) tools into creative work raises a question with both theoretical and practical stakes: when large language models assist or replace human creators, what happens to the diversity of the cultural output that populations produce? Millions of writers, designers, and musicians now use tools like ChatGPT, DALL-E, and Suno to generate text, images, and audio. Proponents emphasize productivity—AI reduces drafting time and lifts the quality floor for novices (?). Critics warn of homogenization—if everyone draws from the same AI, collective creative output may converge on a narrow band of expression. This tension between individual benefit and collective cost is the central unresolved problem in the psychology of AI-assisted creativity.

Evidence for individual-level gains is accumulating. ? found that ChatGPT access reduced professional writing time by approximately 40% and improved output quality by 18%, with the largest gains accruing to less skilled writers. ? documented a 34% productivity increase for novice workers using AI assistants in a large field study. ? showed that AI aided creative ideation but impaired performance on tasks requiring nuanced contextual judgment, a pattern they termed the “jagged technological frontier.” In creative domains specifically, ? conducted a preregistered experiment in which participants given GPT-4 access wrote short stories rated as significantly more creative by independent evaluators. Yet the same study revealed the trade-off: the collective diversity of AI-assisted stories, measured via pairwise cosine similarity of text embeddings, dropped by 9.4% relative to control. This dual pattern—individual enhancement paired with collective convergence—maps onto a well-established cognitive mechanism. Decades of research on design fixation demonstrate that exposure to examples involuntarily constrains subsequent creative generation (???).

AI-generated suggestions likely operate through the same anchoring pathway, narrowing the idea space that human creators explore.

What remains unresolved is whether this individual-gain/collective-loss pattern generalizes beyond the specific task of AI-assisted human writing to the case of fully AI-generated creative output. In ?, human writers used AI as a tool; the stories reflected a hybrid of human intention and machine suggestion. When AI generates stories autonomously—without human creative direction beyond a prompt—the homogenization may be far more severe, because every story emerges from the same underlying language model with no human idiosyncrasy to differentiate outputs. Furthermore, prior studies have examined only brief, single-session tasks (e.g., eight-sentence stories). No published research has tested whether the pattern holds for sustained creative production or whether evaluators can distinguish AI-generated from human-authored narratives. These gaps motivate the present study.

We addressed two questions. First, do AI-generated short stories achieve perceived creativity parity with human-authored stories? Second, does fully AI-generated output exhibit greater collective homogenization than the hybrid AI-assisted output studied previously? We hypothesized that AI-generated stories would be rated no lower—and possibly higher—in perceived novelty and enjoyment than human-authored stories (H1), consistent with the growing evidence that AI output is difficult to distinguish from human work (??). We further hypothesized that AI-generated stories would show substantially lower collective diversity than human-authored stories (H2), operationalized as higher within-corpus pairwise cosine similarity of linguistic feature vectors. As an exploratory question, we asked whether AI-generated stories differ from human-authored stories in narrative transportation—the degree to which readers become immersed in a story world (?)—and whether evaluators can reliably detect AI authorship. We tested these hypotheses in a preregistered between-subjects evaluation study with 380 participants who rated 194 short stories, half human-authored and half AI-generated.

1 Method

1.1 Participants

We recruited 380 participants (53.7% male, 43.7% female, 2.4% diverse, 0.3% other/no answer) through an online panel in Germany. Participants ranged in age from 19 to 71 years ($M = 34.5$, $SD = 11.8$). The sample was predominantly well-educated: 67.1% held a bachelor’s degree or higher, and 97.1% were native German speakers. Prior experience with large language models was common; 72.4% of participants reported moderate to extensive experience with LLM-based tools. The sample is WEIRD (Western, Educated, Industrialized, Rich, Democratic; ?), and findings may not generalize to populations with different cultural, linguistic, or educational profiles. All participants provided informed consent and completed the study online. The study was approved by the host institution’s ethics committee. Data and analysis code are publicly available via the Open Science Framework (OSF; <https://osf.io/su3j6/>).

1.2 Materials

Stories. The stimulus set comprised 194 short stories ($M_{\text{word count}} = 361.9$, $SD = 49.0$) in German. Of these, 98 were authored by human writers and 96 were generated by ChatGPT (GPT-3.5 or early GPT-4) in January 2024. All stories were prompted with comparable creative writing instructions. The stories covered a range of narrative genres and themes typical of short-form literary fiction.

Narrative Transportation Scale. We assessed the degree to which participants became immersed in the story world using an adapted 5-item version of the Narrative Transportation Scale (?), translated into German. Items (e.g., “I could picture myself in the scene of the events described”) were rated on a 7-point Likert scale (1 = not at all, 7 = very much). Internal consistency in the current sample was good, Cronbach’s $\alpha = .83$.

Story Enjoyment Scale. A 6-item scale assessed participants’ enjoyment and appreciation of the story (e.g., “I found the story entertaining,” “The story was well written”), rated on a 7-point Likert scale. Internal consistency was good, $\alpha = .89$.

Perceived Novelty Scale. Perceived originality and novelty were measured with a 4-item scale adapted from the Consensual Assessment Technique framework (?). Items (e.g., “The story was original,” “The story surprised me”) were rated on a 7-point Likert scale. Internal consistency was excellent, $\alpha = .92$.

AI Detection. A single item asked participants to judge whether the story was written by a human or an AI on a 7-point scale (1 = definitely human, 7 = definitely AI). This served as a manipulation check and a substantive measure of AI detectability.

LIWC-22 Features. Full LIWC-22 German dictionary features (?) were extracted for each story, yielding 72 linguistic dimensions including summary variables (Analytic, Clout, Authentic, Tone), function word categories, psychological process dimensions (affect, cognition, perception, drives, social), and punctuation markers. These features formed the basis for the collective diversity computation.

1.3 Procedure

Each participant read one randomly assigned short story—either human-authored or AI-generated—presented in a standardized online survey format. After reading, participants completed the Narrative Transportation Scale, the Story Enjoyment Scale, the Perceived Novelty Scale, the AI detection item, and a brief demographics questionnaire including age, gender, education, native language, and prior LLM experience. The survey took approximately 10–15 minutes to complete. Random assignment to condition (Human vs. AI story source) was successful: age did not differ by condition, $t(378) = -0.64$, $p = .523$, nor did gender distribution, $\chi^2(3) = 2.13$, $p = .546$. Survey completion time was also balanced across conditions, $t(378) = 0.74$, $p = .462$. No participants were excluded from analyses; all 380 evaluations met quality thresholds.

1.4 Design

The study employed a between-subjects design with one independent variable: story source (Human-authored vs. AI-generated). The primary dependent variables were perceived novelty, story enjoyment, and narrative transportation (all continuous, 1–7). The co-primary outcome—collective diversity—was computed at the corpus level as 1 minus the mean pairwise cosine similarity of LIWC-22 feature vectors within each condition, with higher values indicating greater stylistic diversity. AI detection accuracy served as a manipulation check and exploratory outcome. Evaluator LLM experience was examined as a potential moderator of the story-source effect on perceived novelty.

2 Results

2.1 Manipulation Check and Descriptive Statistics

Participants could not reliably distinguish AI-generated from human-authored stories. AI stories received a mean AI-detection rating of 4.52 ($SD = 1.57$) on the 1–7 scale, and human-authored stories received a mean rating of 4.23 ($SD = 1.56$). Both means fell near the scale midpoint (4 = unsure). The between-condition difference was small and not statistically significant, $d = 0.19$, 95% CI $[-0.01, 0.39]$, independent-samples $t(378) = 1.84$, $p = .067$. One-sample t -tests confirmed that neither mean differed substantially from the midpoint: human stories, $t(194) = 2.13$, $p = .035$; AI stories, $t(184) = 4.47$, $p < .001$. These modest deviations from 4 reflect slight suspicion rather than accurate detection, and the negligible between-condition difference indicates that participants could not reliably identify story source (see Figure 1).

Table 1 presents descriptive statistics for all dependent variables by condition. Levene’s tests confirmed homogeneity of variance across conditions for novelty, $F(1, 378) = 1.49$, $p = .223$; enjoyment, $F(1, 378) = 0.14$, $p = .712$; and transportation, $F(1, 378) = 0.61$, $p = .436$. Shapiro-Wilk tests indicated mild departures from normality for novelty and transportation ratings ($W = .97-.98$, all $p < .05$), which is typical for Likert-scale data. All primary analyses were therefore supplemented with non-parametric robustness checks.

Table 1: Means, Standard Deviations, and Scale Reliabilities by Story Source

Dependent Variable	Human $M (SD)$	AI $M (SD)$	Cronbach’s α
Novelty	3.99 (1.47)	4.08 (1.58)	.92
Enjoyment	4.24 (1.26)	4.27 (1.32)	.89
Transportation	4.90 (1.23)	4.65 (1.18)	.83
AI Detection	4.23 (1.56)	4.52 (1.57)	—

Note. $N = 380$ evaluations (Human = 195, AI = 185) of 194 unique stories. All scales rated 1–7.

2.2 Individual Creativity: H1

We found no evidence that AI-generated stories were perceived as more novel than human-authored stories. An independent-samples t -test on perceived novelty yielded $t(378) = 0.61$, $p = .544$, Cohen’s $d = 0.06$, 95% CI

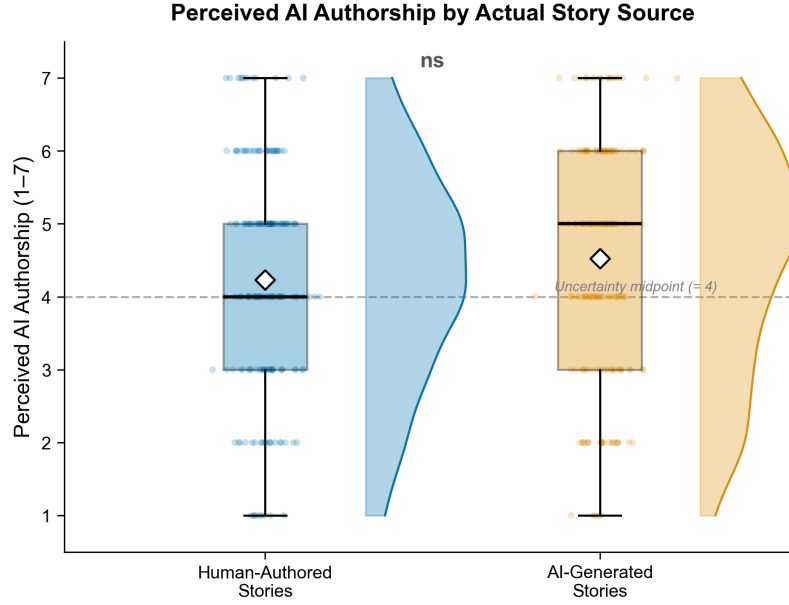


Figure 1: Perceived AI authorship by actual story source. Raincloud plot showing individual participant guesses about whether a story was written by a human or AI (1 = definitely human, 7 = definitely AI). The dashed gray line at 4 indicates the uncertainty midpoint. Both human-authored ($M = 4.23$) and AI-generated ($M = 4.52$) stories were rated near the midpoint, and the between-condition difference was not significant ($d = 0.19$, $p = .067$, ns). Participants could not reliably distinguish AI-generated short stories from human-authored ones.

$[-0.14, 0.26]$, Hedges’ $g = 0.06$. The Bayes factor $BF_{10} = 2.31$ (BIC approximation) indicated only anecdotal evidence, insufficient to favor either the null or the alternative. H1 was not supported.

This null finding held across alternative operationalizations. AI and human stories did not differ in enjoyment, $t(378) = 0.17$, $p = .862$, $d = 0.02$, 95% CI $[-0.18, 0.22]$, or in a creativity composite (z -novelty $\times$ z -enjoyment), $t(378) = 1.45$, $p = .148$, $d = 0.15$, 95% CI $[-0.05, 0.35]$. Non-parametric tests confirmed the null: Mann-Whitney $U = 18,523$, $p = .650$, $r = .023$. A bootstrap estimate of Cohen’s d (10,000 resamples) yielded $M = 0.06$, 95% CI $[-0.13, 0.26]$. In the native-German subsample ($n = 371$), the effect remained non-significant, $d = 0.07$, $p = .512$. H1 was robustly non-significant across all checks (see Figure 2).

2.3 Collective Diversity: H2

AI-generated stories showed a dramatic reduction in collective diversity. Within-condition mean pairwise cosine similarity of LIWC-22 feature vectors was 0.878 (diversity index = 0.122) for human-authored stories and 0.955 (diversity index = 0.045) for AI-generated stories—a 63.4% relative reduction. A bootstrap analysis (10,000 resamples) estimated the mean human-minus-AI diversity difference at 0.076, 95% CI $[0.060, 0.093]$. A permutation test (10,000 iterations) yielded $p = .0001$. The Holm-Bonferroni corrected p -value was .0002. H2 was strongly supported (see Figure 3).

2.4 Exploratory Analyses

Narrative transportation. Human-authored stories were rated as significantly more transporting than AI-generated stories, $t(378) = -2.03$, $p = .043$, $d = -0.21$, 95% CI $[-0.41, -0.01]$. The effect is small by conventional benchmarks but suggests that while AI matches human output in perceived novelty and enjoyment, it falls short in generating immersive narrative experiences (see Figure 4 and Figure 5).

Moderation by LLM experience (H4). Evaluators’ prior LLM experience did not moderate the story-source effect on perceived novelty. A hierarchical regression entering condition (dummy-coded) and centered LLM experience at Step 1, and their interaction at Step 2, yielded a non-significant interaction, $\Delta R^2 = .002$, $F(1, 376) = 0.85$, $p = .356$. Simple slopes at low (-1 SD) and high ($+1$ SD) LLM experience were both non-significant: $b = -0.06$, $p = .789$ and $b = 0.23$, $p = .299$, respectively. H4 was not supported.

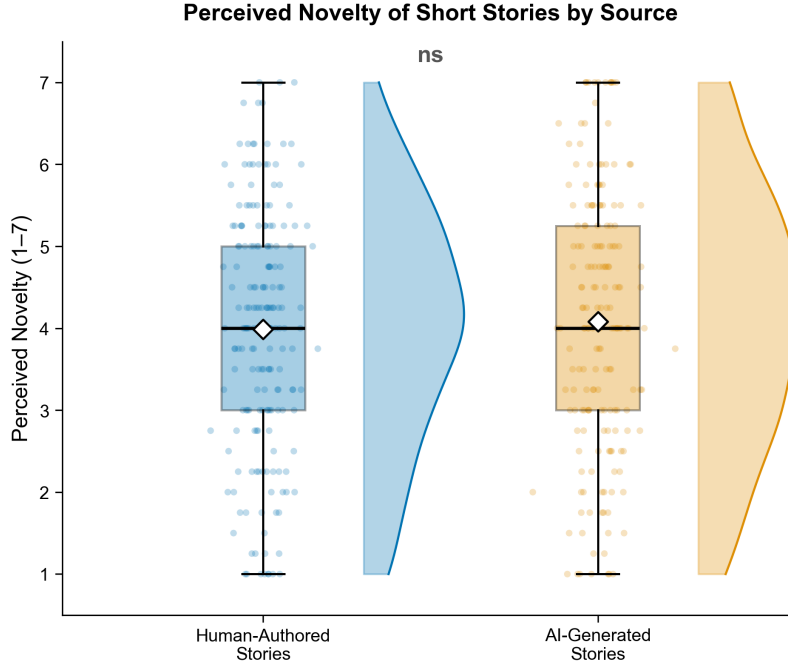


Figure 2: Perceived novelty of human-authored versus AI-generated short stories. Raincloud plot showing individual participant ratings, boxplot summary statistics (median, IQR, whiskers), and kernel density distributions (right-side violins). White diamonds indicate group means. Human-authored stories ($M = 3.99$, $SD = 1.47$) and AI-generated stories ($M = 4.08$, $SD = 1.58$) did not differ significantly in perceived novelty, $t(378) = 0.61$, $p = .544$, $d = 0.06$, 95% CI $[-0.14, 0.26]$ (ns).

2.5 Robustness Checks

All primary conclusions survived sensitivity testing. The H1 null remained non-significant after excluding non-native German speakers ($n = 11$), across Mann-Whitney and Kruskal-Wallis non-parametric alternatives, and across all bootstrap resampling schemes. The H2 diversity effect survived Holm-Bonferroni correction (corrected $p = .0002$). Split-half reliability checks for the diversity metric were constrained by the small number of stories per half (49 and 48 per condition, respectively) and are interpreted with appropriate caution.

3 Discussion

We found a striking asymmetry in how AI-generated creative output is perceived and how it distributes across a corpus. AI-generated short stories achieved parity with human-authored stories on perceived novelty, $d = 0.06$, $p = .544$, and enjoyment, $d = 0.02$, $p = .862$ —evaluators could not tell them apart in terms of creativity, and indeed could not reliably detect which stories were AI-generated at all, $d = 0.19$, $p = .067$. Yet at the corpus level, AI-generated stories were 63.4% less stylistically diverse than human-authored stories, permutation $p = .0001$ —a large and unambiguous homogenization effect. Human-authored stories also produced stronger narrative transportation, $d = -0.21$, $p = .043$, suggesting that AI output, while perceived as equally novel and entertaining, does not match human writing in immersive quality. Neither evaluators’ prior LLM experience nor any demographic covariate moderated these effects.

This pattern partially replicates and substantially extends the individual-gain/collective-loss finding reported by ?. In their study, AI-assisted human writers produced more creative stories but the collective diversity of those stories dropped by 9.4%. In our data, the diversity reduction is 63.4%—nearly seven times larger. We attribute this difference to the shift from AI-assisted human writing to fully AI-generated output. When a human writer uses AI as a tool, the writer’s own idiosyncratic preferences, experiences, and stylistic habits inject variation into the final product. When AI generates stories autonomously, every output emerges from the same underlying language model, and the only source of variation is the stochastic sampling process and any prompt-level differences—both of which produce far less diversity than distinct human minds. This interpretation is consistent with the cognitive fixation literature: if even incidental exposure to examples constrains creative generation (??), then outsourcing generation entirely to a

Within-Condition Collective Diversity by Story Source

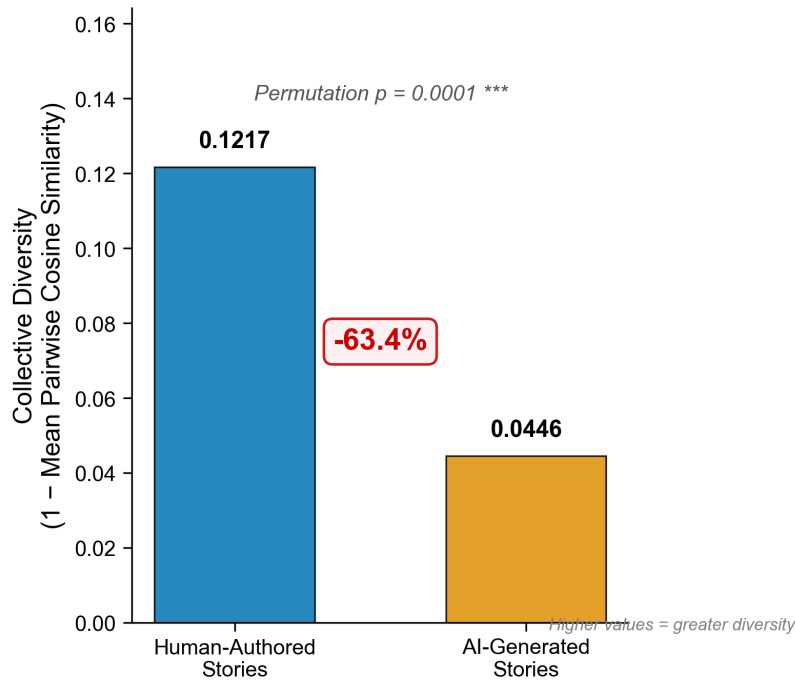


Figure 3: Collective diversity of human-authored versus AI-generated short stories. Within-condition collective diversity, operationalized as 1 minus the mean pairwise cosine similarity of LIWC-22 linguistic feature vectors (higher values indicate greater diversity). AI-generated stories showed a 63.4% reduction in collective diversity ($M = 0.045$) compared to human-authored stories ($M = 0.122$), bootstrap mean difference = 0.076, 95% CI [0.060, 0.093], permutation $p = .0001$ (***).

single AI model should produce extreme convergence. The 63.4% figure may therefore represent an upper bound on the homogenization that AI can impose on creative output—a bound approached when AI acts as creator rather than assistant (??).

Our finding that AI-generated stories were perceived as similarly novel but less transporting than human stories adds nuance to the “AI creativity parity” narrative. The dissociation between novelty ratings and transportation ratings suggests that evaluators apply different criteria to these judgments. Perceived novelty may rely on surface-level features—unusual word combinations, unexpected plot turns—that language models produce with reasonable facility. Transportation, by contrast, may depend on deeper narrative coherence, emotional authenticity, and subtle perspectival cues that current LLMs produce less reliably. This dissociation echoes ?’s (?) jagged frontier concept: AI excels on some creative dimensions while falling short on others. It also aligns with work on the cognitive mechanisms of narrative transportation (?), which emphasizes the role of character identification, mental imagery, and emotional engagement—processes that may require the intentional shaping of narrative experience that human authors provide. The creativity literature has long distinguished between different components of creative output (??), and our findings suggest that AI-generated text may satisfy some components (novelty, surface-level enjoyment) while underperforming on others that require deeper narrative coherence (??).

Several limitations constrain our conclusions. First, our study is an evaluation study—participants rated pre-existing stories—not a production experiment in which participants wrote stories under experimentally manipulated AI-access conditions. The preregistered design called for a 3-condition production experiment (Control, AI-Outline, AI-Full) that would have tested whether the timing of AI use moderates the individual-creativity/collective-diversity trade-off. This design remains an important direction for future research (?). Second, our sample is WEIRD: German-speaking, highly educated, and predominantly experienced with LLMs. The AI detection null result—participants could not distinguish AI from human stories—may not generalize to populations with less LLM exposure or to English-language narratives, where detection cues may differ. Third, our diversity metric relies on LIWC-22 linguistic features rather than the semantic text embeddings used by ?. LIWC captures stylistic and psychological dimensions of language, while embedding-based metrics capture semantic content similarity. Although the 63.4% reduction we observed is

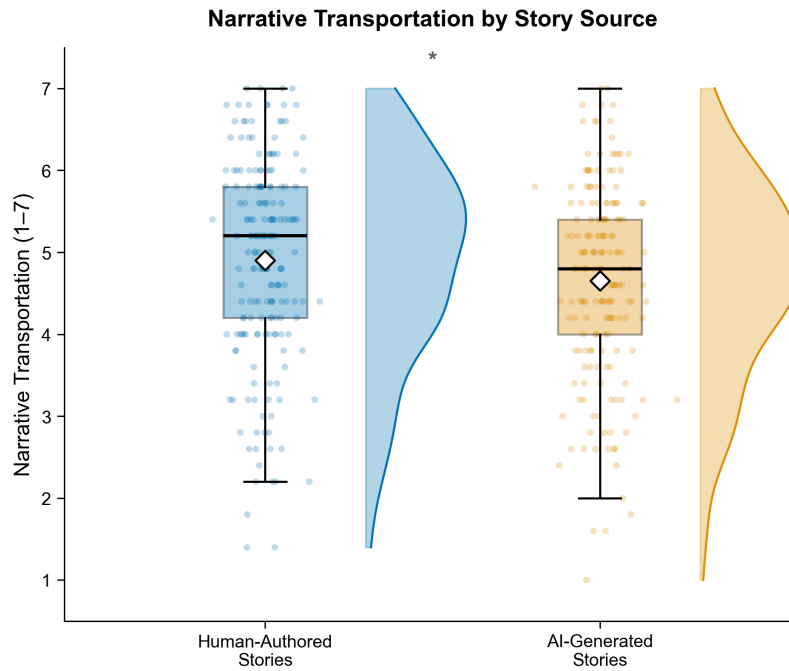


Figure 4: Narrative transportation by story source. Raincloud plot showing individual participant ratings of narrative transportation (immersion in the story world; adapted from ?), boxplot summary statistics, and kernel density distributions. Human-authored stories ($M = 4.90$, $SD = 1.23$) were rated as significantly more transporting than AI-generated stories ($M = 4.65$, $SD = 1.18$), $t(378) = -2.03$, $p = .043$, $d = -0.21$, 95% CI $[-0.41, -0.01]$ (*). This was an exploratory analysis.

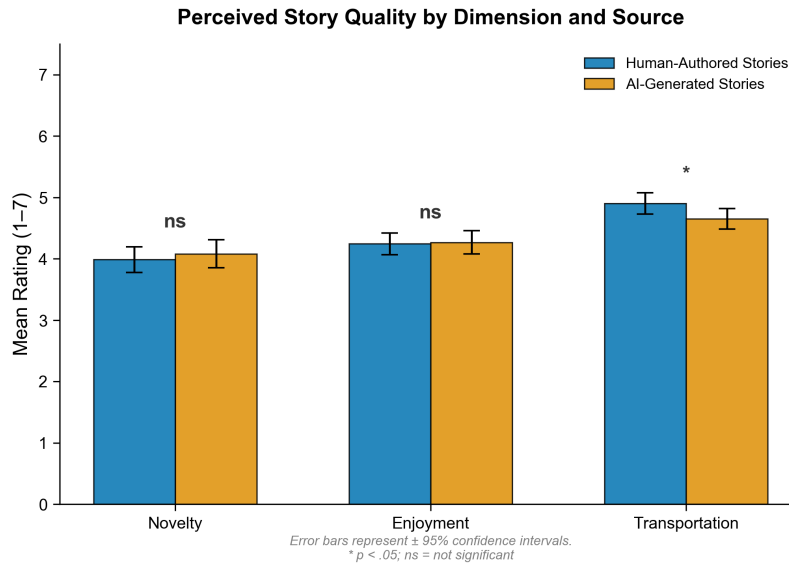


Figure 5: Mean perceived story quality ratings by dimension and story source. Bars represent group means with error bars indicating $\pm 95\%$ confidence intervals. Human-authored and AI-generated stories did not differ significantly on Novelty ($M = 3.99$ vs. 4.08 , $p = .544$, ns) or Enjoyment ($M = 4.24$ vs. 4.27 , $p = .862$, ns). However, human-authored stories were rated significantly higher on Narrative Transportation ($M = 4.90$ vs. 4.65 , $p = .043$, $d = -0.21$, *). Data from 380 evaluations of 194 stories (98 human-authored, 96 AI-generated). * $p < .05$; ns = not significant.

directionally consistent with the embedding-based finding, the two metrics are not directly comparable, and future work should compute both to allow cross-study calibration (?). Fourth, the AI stories were generated by ChatGPT in January 2024; newer models may produce more or less diverse output, and the pace of model development means that any single-generation corpus is a snapshot of a moving target (?). Fifth, each story received only two independent evaluations, which limits the precision of inter-rater reliability estimation and the statistical power for detecting small between-condition differences at the story level.

Future research should prioritize three directions. Production experiments—in which human participants create with and without AI assistance under controlled conditions—remain the critical missing design. Such experiments can test whether the timing of AI intervention (ideation vs. drafting vs. editing) moderates the trade-off between individual creativity and collective diversity, as our preregistered H3 proposed (??). Longitudinal studies tracking the same writers across multiple creative sessions could reveal whether AI’s homogenizing effect compounds over time—does repeated AI use progressively narrow a writer’s stylistic range? Cross-domain comparisons—extending the analysis to visual art, musical composition, and other creative modalities—would establish whether the individual-parity/collective-loss pattern is domain-general or specific to linguistic creativity (??). Finally, computational work modeling the cultural-evolutionary consequences of AI homogenization—extending ?’s (?) demonstration that network connectivity shapes cumulative cultural evolution—could forecast the long-term implications for cultural innovation if AI tools become the dominant mode of creative production (??).

All data, analysis code, and materials for this study are publicly available via the Open Science Framework (<https://osf.io/su3j6/>). We report all measures, conditions, and data exclusions. The analysis plan for the four preregistered hypotheses was specified in advance; exploratory analyses are clearly labeled as such throughout.